A Local Stochastic Volatility Model for the CAC40: Particle Calibration and Model Risk on Barriers and Autocalls

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October 2026

Abstract

We implement a Local Stochastic Volatility (LSV) model on the CAC40 of 12 February 2025: the spot follows a local volatility scaled by the square root of a Heston variance, $\mathrm{d}S/S = \mu\,\mathrm{d}t + L(t,S)\sqrt{v_t}\,\mathrm{d}W$, and the leverage function L is calibrated with the particle method of Guyon and Henry-Labordère so that the model reprices the Dupire local volatility of an SSVI fit of the market surface. A mixing fraction $\eta \in [0,1]$ scales the volatility of variance: $\eta = 0$ is the pure local volatility model, $\eta = 1$ keeps the vol-of-vol of the calibrated Heston model. The calibration takes 13 seconds with 50,000 particles on a weekly grid up to 4.86 years; the model reprices the 142 market quotes with an implied volatility RMSE of 0.25 vol point (0.43 for Heston), of which 0.09 is numerical error with respect to its SSVI target. At identical vanilla prices, the up-and-out call (strike 8000, barrier 8800, one year) is worth 34.6 in local volatility and 64.7 in LSV with $\eta = 1$, the down-and-out call 423.3 and 408.5, and the fair coupon of a 4-year Phoenix moves from 7.31% to 6.26% a year, while the Athena coupon moves by 0.2 point only. These differences are model risk: vanillas do not identify η.

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1 Introduction

Two families of models dominate equity derivatives pricing. *Local volatility* [4] reprices every vanilla option exactly, but its dynamics are poor: the forward smile flattens and the volatility is a deterministic function of the spot. *Stochastic volatility*, such as Heston [5], has realistic dynamics but only a few parameters: on the CAC40 surface of 12 February 2025, the calibrated Heston model of [1] leaves an implied volatility RMSE of 0.42 vol point, with errors up to 0.8 point on the long maturities.

The *Local Stochastic Volatility* model combines both [10, 11]: a leverage function $L(t, S)$ multiplies a stochastic volatility $\sqrt{v_t}$ and is chosen so that the model reprices the market. It is the market standard for path-dependent equity products such as barriers and autocalls [12]. The share of volatility carried by the stochastic part, governed here by a mixing fraction η, is not identified by vanillas: all values of η give the same vanilla prices but different exotic prices.

This report documents the implementation (Section 3), checks the calibration (Section 4) and measures the model risk on the barrier options and autocallable notes of the companion reports [2, 3] (Section 5).

2 Model

2.1 Forward log-moneyness

Rates and dividends are deterministic and enter through the forward curve $F(0, t)$ bootstrapped from put–call parity [1]. The state variable is the forward log-moneyness $y_t = \ln(S_t/F(0, t))$:

$$\mathrm{d}y_t = -\tfrac{1}{2}\sigma_t^2\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t^S, \qquad \mathbb{E}[S_t] = F(0, t), \tag{1}$$

where σ_t is the instantaneous volatility of the spot. Implied volatilities are functions of (y, T) with $y = \ln(K/F(0, T))$, and $w(y, T) = \sigma_{\text{imp}}^2(y, T)\,T$ is the total implied variance.

2.2 Dupire local volatility

The local volatility model $\sigma_t = \sigma_{\text{loc}}(t, y_t)$ reprices the implied surface if [4, 6]

$$\sigma_{\text{loc}}^2(T, y) = \frac{\partial_T w}{1 - \frac{y}{w}\partial_y w + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)(\partial_y w)^2 + \frac{1}{2}\partial_{yy} w}. \tag{2}$$

2.3 Local stochastic volatility

$$\sigma_t = L(t, y_t)\sqrt{v_t}, \qquad \mathrm{d}v_t = \kappa(\theta - v_t)\,\mathrm{d}t + \eta\,\xi\sqrt{v_t}\,\mathrm{d}W_t^v, \qquad \mathrm{d}\langle W^S, W^v\rangle_t = \rho\,\mathrm{d}t, \tag{3}$$

with $(v_0, \theta, \kappa, \xi, \rho)$ the parameters of the calibrated Heston model (Table 1) and $\eta \in [0, 1]$ the mixing fraction. By Gyöngy's theorem [8], the one-dimensional marginals of y_t are those of the local volatility model, hence the vanilla prices are the same, if and only if

$$L^2(t, y)\,\mathbb{E}[v_t \mid y_t = y] = \sigma_{\text{loc}}^2(t, y). \tag{4}$$

For $\eta = 0$ the variance is deterministic, $\mathbb{E}[v_t \mid y_t] = v_t$, and the model is the local volatility model. For $\eta = 1$ and $L \equiv 1$ it is the Heston model.

3 Numerical Method

3.1 Smooth implied surface

Equation (2) needs a surface smooth in y and T and free of arbitrage. The 142 quotes (13 expiries from 9 days to 4.86 years) are fitted by:

v_0	θ	κ	ξ	ρ	Heston IV RMSE
0.0197	0.0438	1.308	0.482	-0.605	0.421 vol pts

Table 1: Stochastic variance parameters: the Heston model calibrated on the CAC40 surface [1].

- **SSVI** [7]: $w(y,T) = \frac{\theta_T}{2}\big(1+\rho_s\varphi y+\sqrt{(\varphi y+\rho_s)^2+1-\rho_s^2}\big)$, $\varphi = \eta_s/(\theta_T^{\gamma}(1+\theta_T)^{1-\gamma})$, with the market ATM total variance θ_T (non-decreasing) and three global parameters $\rho_s = -0.454$, $\eta_s = 1.376$, $\gamma = 0.418$, chosen in the region free of butterfly arbitrage; RMSE 0.218 vol point;
- **SVI**: one 5-parameter smile per expiry, total variance linear in T at fixed y between expiries; RMSE between 0.003 and 0.079 vol point per expiry.

SSVI is the default: its fit is coarser, but its slices never cross and its local volatility is regular in maturity, whereas the SVI slices of the long maturities are unstable.

3.2 Local volatility

The derivatives of (2) are central finite differences of the fitted surface ($\Delta y = 10^{-3}$, $\Delta T = 10^{-4}$). The denominator is floored at 10^{-3} and the local volatility clipped to $[3\%, 150\%]$ to stay finite in the far wings.

3.3 Particle calibration of the leverage

The conditional expectation of (4) is unknown: it is estimated on the simulation itself (Algorithm 1, [9]).

Algorithm 1 Particle calibration of the leverage function

N particles $y^{(n)} = 0$, $v^{(n)} = v_0$; time grid $t_i = i\,\Delta t$, $\Delta t = 1/52$, up to $T_{\max} = 4.86$
for $i = 0, \dots, M$ **do**
  grid $\{g_j\}_{j=1}^{60}$ between the 0.05% and 99.95% quantiles of the $y^{(n)}$
  $h = 1.5\,\mathrm{std}(y)\,N^{-1/5}$, $\quad m_j = \dfrac{\sum_n K_h(y^{(n)} - g_j)\,v^{(n)+}}{\sum_n K_h(y^{(n)} - g_j)}$ ▷ Gaussian kernel
  $L(t_i, g_j) = \sigma_{\mathrm{loc}}(t_i, g_j)/\sqrt{m_j}$
  move every particle by one Euler step with $\sigma^{(n)} = L(t_i, y^{(n)})\sqrt{v^{(n)+}}$ (linear interpolation in y)
end for

At $t = 0$ all particles sit at $y = 0$ and $\mathbb{E}[v_0 \mid y] = v_0$. For $\eta = 0$ the regression is not needed. The variance follows an Euler full truncation scheme. The leverage is stored on the calibration grid and read at the last calibration date before t by the pricing simulation, which uses independent paths (no reuse of the particles, hence no in-sample bias).

3.4 Pricing

`LSVModel.generate_paths` returns the spot and its instantaneous variance $L^2 v$, the interface of the Heston model: the Monte Carlo pricer, the Greeks and every product of the companion reports apply unchanged. The continuous barriers use the Brownian bridge with the variance $L^2 v$ of each step [2].

3.5 Cost

The calibration costs $\mathcal{O}(MNG)$ in time for the kernel regressions ($M = 253$ steps, N particles, $G = 60$ grid points) and $\mathcal{O}(NG)$ in memory for the kernel weights of one step, the particles not being stored along time; the leverage itself takes $\mathcal{O}(MG)$. Measured times: 0.6 s for $\eta = 0$, 13 s for $\eta = 1$ with $N = 50{,}000$ (Table 4); about 7 s to reprice the 142 quotes on 100,000 daily paths up to 4.86 years. A Vega or Volga of an LSV model recalibrates the leverage twice.

4 Calibration Results

4.1 Local volatility

Table 2 and Figure 1 show the SSVI local volatility. The local skew is about twice the implied skew, and the short-dated wings exceed 40% at -30% of moneyness.

$T \setminus y$	-30%	-20%	-10%	0%	$+10\%$	$+20\%$
0.1	43.90	36.13	26.29	13.36	15.04	21.29
0.5	34.74	28.95	22.02	14.61	12.76	16.30
1	29.89	25.16	19.78	14.47	12.22	13.99
2	28.79	24.68	20.29	16.21	13.84	14.08
4	25.65	22.52	19.35	16.48	14.49	13.86

Table 2: Dupire local volatility (%) of the SSVI surface.

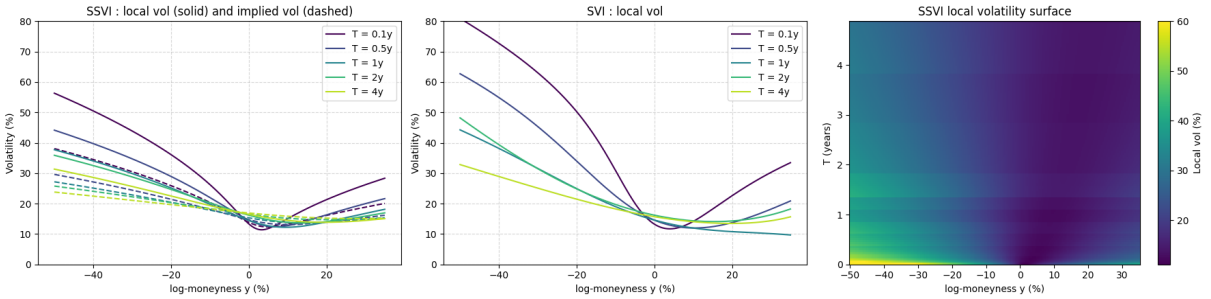


Figure 1: Left: SSVI local (solid) and implied (dashed) volatilities. Middle: SVI local volatility. Right: SSVI local volatility surface.

4.2 Leverage function

Figure 2 shows the calibrated leverage. With $\eta = 1$, beyond six months it stays between 0.8 and 1.2 for $|y| \leq 20\%$: the Heston model already explains most of the surface, and L corrects its residual errors. Short-dated leverage is larger in the wings, where the local volatility is steep; outside the particle quantiles it is extrapolated flat. With $\eta = 0.5$ the stochastic part carries less of the smile and the leverage is more convex.

4.3 Gyöngy condition

On 100,000 independent pricing paths, the conditional expectation of the spot variance $\mathbb{E}[L^2 v_t \mid y_t = y]$, estimated by kernel regression between the 2% and 98% quantiles of y_t, is compared with σ^2_{loc} (Figure 3): the maximum gap on this range is 0.60, 0.58, 0.93 and 1.81 vol points at $t = 0.25$, 1, 2 and 4 years, the largest gaps being in the wings where few particles lie.

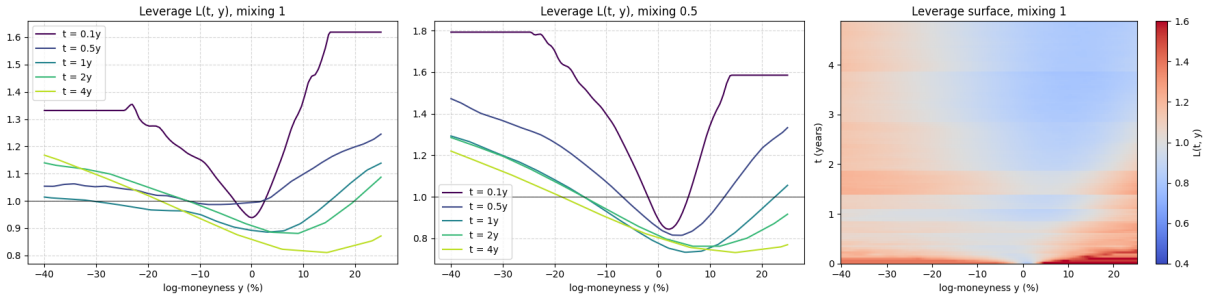


Figure 2: Calibrated leverage $L(t, y)$ for $\eta = 1$ and $\eta = 0.5$, and leverage surface for $\eta = 1$.

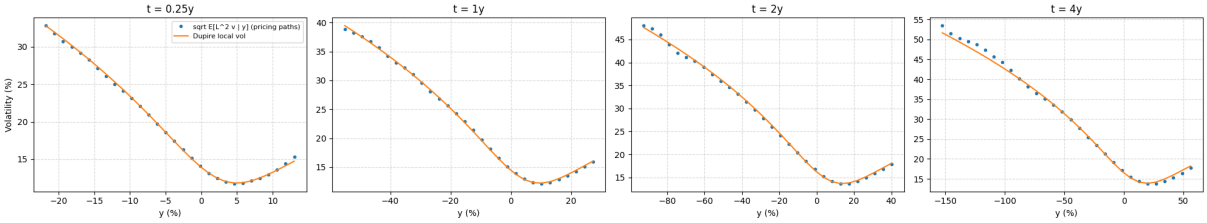


Figure 3: Gyöngy check: $\sqrt{\mathbb{E}[L^2 v_t \mid y]}$ on the pricing paths (dots) and Dupire local volatility (line).

4.4 Repricing of the market

Table 3 gives the implied volatility RMSE of each model on the market quotes (100,000 paths, daily steps). Local volatility and LSV divide the Heston error by 1.7 over the whole surface, and by 1.8 to 3.5 on the three longest maturities, where Heston is the least accurate. The remaining error is mostly the SSVI fit error: the RMSE with respect to the SSVI target is 0.088 vol point (Table 4). The 9-day expiry is the least accurate for the LSV models (0.26 and 0.41 vol point), since the leverage is calibrated on a weekly grid.

Expiry (y)	**Heston**	**Local vol**	**LSV** $\eta = 0.5$	**LSV** $\eta = 1$
0.025	0.681	0.138	0.263	0.414
0.101	0.132	0.216	0.239	0.231
0.178	0.122	0.214	0.239	0.179
0.351	0.175	0.187	0.193	0.083
0.600	0.161	0.203	0.216	0.119
0.849	0.227	0.137	0.168	0.076
1.099	0.097	0.217	0.217	0.187
1.348	0.210	0.339	0.331	0.334
1.597	0.206	0.110	0.129	0.113
1.847	0.248	0.119	0.130	0.140
2.844	0.634	0.225	0.237	0.232
3.841	0.786	0.430	0.444	0.457
4.858	0.799	0.237	0.221	0.227
All quotes	0.428	0.230	0.246	0.245

Table 3: Implied volatility RMSE on the 142 market quotes (vol points, Monte Carlo).

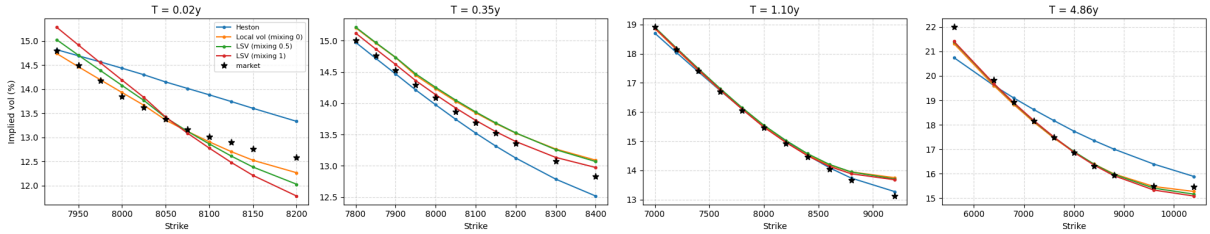


Figure 4: Market smiles (stars) and smiles of the four models at 9 days, 4 months, 1.1 years and 4.9 years.

4.5 Convergence

Table 4 shows the effect of the number of particles for $\eta = 1$. The time is linear in N; the numerical error decreases from 0.147 to 0.083 vol point between 10,000 and 100,000 particles, and 50,000 particles are within 0.005 point of the 100,000 result.

Particles	Calibration time (s)	RMSE vs SSVI	RMSE vs market
10,000	2.3	0.147	0.251
25,000	6.3	0.108	0.246
50,000	13.0	0.088	0.245
100,000	26.5	0.083	0.242

Table 4: Particle calibration, $\eta = 1$ (vol points).

5 Model Risk

Six models price the same products: the Heston model and the LSV model with $\eta \in \{0, 0.25, 0.5, 0.75, 1\}$, all calibrated to the same surface. Products: 1-year European barrier options of strike 8000 (continuous barriers, daily steps) and the 4-year notes of [3] (Athena: annual observations, snowball coupon of 8%, autocall at 100%, protection at 60%; Phoenix: quarterly, coupon barrier at 70%, memory). 50,000 paths with a common seed, without control variate (not available for the local volatility models).

Product	Heston	$\eta = 0$	$\eta = 0.25$	$\eta = 0.5$	$\eta = 0.75$	$\eta = 1$	LSV range	Std error
Put 1y ATM (vanilla)	467.83	470.82	471.82	472.47	472.44	470.62	1.85	3.36
Call 1y ATM (vanilla)	491.03	495.69	496.50	496.73	496.14	494.28	2.45	2.53
PDO, $H = 7200$	16.41	17.58	17.04	16.72	16.42	16.07	1.51	0.33
PDI, $H = 7200$	451.42	453.24	454.78	455.74	456.02	454.55	2.78	3.42
CDO, $H = 7200$	407.54	423.33	423.78	421.27	416.17	408.47	15.32	2.49
CUO, $H = 8800$	63.04	34.58	37.18	44.28	54.74	64.73	30.15	0.67
Down-and-out digital (100), $H = 7200$	44.44	42.90	42.92	43.10	43.60	44.20	1.30	0.17
Athena, European protection	98.57	98.18	98.24	98.33	98.37	98.47	0.29	0.10
fair coupon (% p.a.)	9.44	9.72	9.68	9.61	9.60	9.52	0.20	
Athena, American protection	96.81	96.54	96.55	96.63	96.65	96.77	0.23	0.10
fair coupon (% p.a.)	11.21	11.27	11.30	11.26	11.29	11.20	0.10	
Phoenix	99.95	98.99	99.17	99.38	99.57	99.78	0.79	0.07
fair coupon (% p.a.)	6.06	7.31	7.06	6.77	6.53	6.26	1.05	

Table 5: Prices under the six models (nominal 100 for the notes). LSV range: maximum minus minimum over the five LSV models. Std error: largest Monte Carlo standard error of the row.

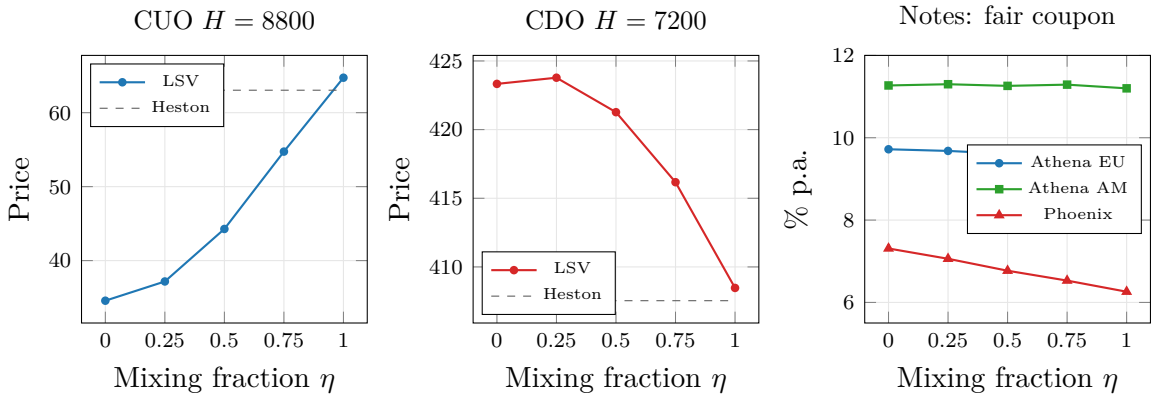


Figure 5: Exotic prices as functions of the mixing fraction, at identical vanilla prices (dashed: Heston).

Vanillas. The vanilla prices of the five LSV models differ by at most 1.85 (put) and 2.45 (call), below the Monte Carlo standard errors (3.36 and 2.53): the mixing fraction does not change them. They are 3 to 6 above the Heston prices, which carry the 0.42 vol point calibration error of Heston.

Barriers. The up-and-out call almost doubles from $\eta = 0$ to $\eta = 1$ (34.6 to 64.7, 45 standard errors), and the down-and-out call loses 14.9 (6 standard errors). The value of the up-and-out call comes from paths that end between the strike and the barrier without touching it; a stochastic volatility makes low-variance scenarios more likely, which raises this probability. Figure 6 shows that the gap holds for every barrier level, reaching 44.6 at $H = 9150$ between local volatility and LSV $\eta = 1$. The down-and-out put and the down-and-out digital move by 1.5 and 1.3 (5 and 8 standard errors): far barriers on the downside depend less on the dynamics.

Autocalls. The Athena is almost insensitive to η (fair coupon 9.72% to 9.52%, 0.10 point with an American protection): it depends mainly on the terminal distributions at the observation dates, which every LSV model shares, and on the probability of a call at the first date. The Phoenix, whose coupons depend on the path above the coupon barrier at every quarter, moves by 1.05 point of coupon (7.31% to 6.26%).

Heston. Heston is close to LSV $\eta = 1$ on the barriers (CUO 63.0 against 64.7, CDO 407.5 against 408.5), as expected from a leverage close to 1, but its notes inherit its vanilla errors (Phoenix fair coupon 6.06% against 6.26%).

6 Greeks

Table 6 compares the Greeks of the up-and-out call and of the Athena (common random numbers, 50,000 paths). For the LSV models, Vega and Volga shift the whole implied surface by one vol point and recalibrate the leverage, the market definition of a Vega; for Heston they shift $\sqrt{v_0}$ and $\sqrt{\theta}$, so the two are not directly comparable. Delta keeps the surface fixed in moneyness.

The Greeks of the Athena are nearly model independent (Vega -0.53 to -0.58, Dividend -1.04 to -1.11), like its price. The up-and-out call has a negative Vega in every model, larger in absolute value when the volatility is more stochastic, and a Gamma four times larger with $\eta = 1$ than in local volatility: hedging it with a local volatility model would underestimate both.

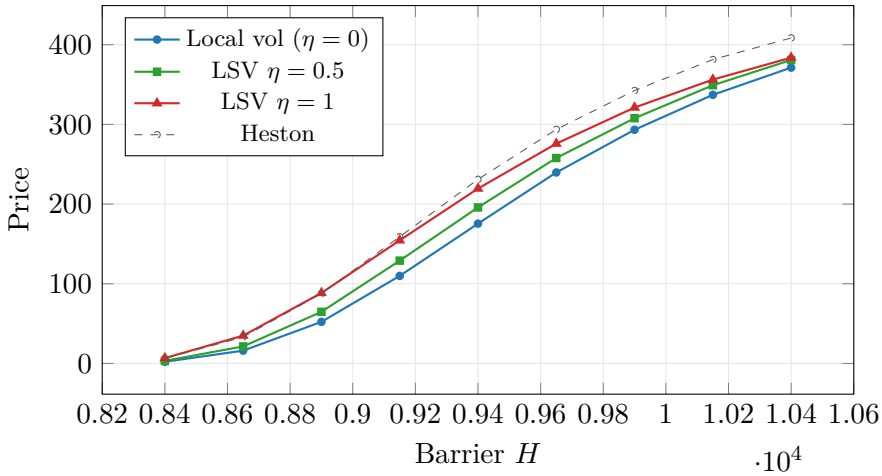


Figure 6: Up-and-out call ($K = 8000$, $T = 1$) as a function of the barrier.

	CUO 1y, $H = 8800$				**Athena 4y, European protection**			
	$\eta = 0$	$\eta = 0.5$	$\eta = 1$	Heston	$\eta = 0$	$\eta = 0.5$	$\eta = 1$	Heston
Price	34.58	44.28	64.73	63.04	98.18	98.33	98.47	98.57
Delta (per point)	−0.0457	−0.0547	−0.0491	−0.0618	0.00481	0.00491	0.00496	0.00505
Gamma ($\times 10^5$)	−5	−8	−23	−21	−1	−1	−1	0
Vega (per vol pt)	−6.05	−7.03	−7.94	−11.46	−0.575	−0.557	−0.527	−0.534
Volga	1.45	1.71	1.16	1.79	−0.035	−0.002	−0.045	−0.040
Rho (per 1%)	−1.24	−1.53	−1.97	−1.45	−0.743	−0.803	−0.748	−0.774
Dividend (per 1%)	0.89	1.08	1.33	0.82	−1.086	−1.038	−1.114	−1.111
Theta (per day)	0.107	0.173	0.188	0.210	0.0086	0.0063	0.0052	0.0049

Table 6: Greeks under local volatility, LSV and Heston (168 s in total, most of it for the recalibrations of the Vegas).

7 Implementation

- `Model/local_vol.py`: `LocalVolSurface` (Dupire from SSVI, SVI or a flat volatility; implied surface shift for the Vega), `LSVModel` (particle calibration, `generate_paths`, `leverage`, `vol_shifted`) and `reprice_surface`;
- `Produits/market.py`: `cac40_lsv(mixing)` builds a calibrated LSV model on the CAC40;
- `Model/lsv_test.ipynb`: the notebook of this report (6 to 8 minutes).

```
from market import cac40_heston, cac40_lsv
from products import BarrierOption, Phoenix
from monte_carlo import MonteCarloPricer

mkt = cac40_heston()
lsv = cac40_lsv(mkt, mixing=0.5)                    # Dupire (SSVI) + leverage
    , 50,000 particles
pricer = MonteCarloPricer(n_paths=50_000, steps_per_year=252, seed=42,
    control_variate=False)
print(pricer.price(lsv, BarrierOption.from_code("CUO", 8000, 8800, 1.0)))
note = Phoenix(4.0, mkt.S0, 0.06, coupon_barrier=0.7, frequency=0.25)
print(note.fair_coupon(MonteCarloPricer(n_paths=50_000, steps_per_year=52, seed
    =42, control_variate=False), lsv))
```

8 Limitations and Next Steps

- *Mixing fraction.* Vanillas do not identify η; it must be calibrated to exotic quotes (barriers, forward-start options, VIX-type products), not available here. The model risk of Table 5 is the range of prices compatible with the vanilla surface.
- *Implied surface.* The SSVI fit error (0.22 vol point) is inherited by every model; a richer arbitrage-free parametrisation (e.g. eSSVI) would lower it.
- *Short maturities.* The weekly calibration grid limits the accuracy of the 9-day expiry (0.41 vol point for $\eta = 1$); a finer grid at the start would fix it.
- *Particle method.* Kernel bias in the wings (Gyöngy gaps up to 1.8 vol points at 4 years beyond the central quantiles), Euler scheme with the Feller condition violated; a PDE calibration (Fokker–Planck) is the standard alternative.
- *Dynamics.* Sticky-moneyness spot bumps; one stochastic factor; no stochastic rates or dividends.

9 Conclusion

A Local Stochastic Volatility model calibrated by the particle method reprices the CAC40 surface with an implied volatility RMSE of 0.25 vol point, against 0.43 for Heston, in 13 seconds. At identical vanilla prices, the mixing fraction moves the up-and-out call by a factor of two, the down-and-out call by 15 and the Phoenix fair coupon by one point of coupon, while the Athena moves by 0.2 point: path-dependent products whose value depends on the path between observation dates or near a barrier carry a model risk that vanilla calibration alone cannot remove.

A Notation

Symbol	Meaning
$y = \ln(S/F(0,t))$	Forward log-moneyness
$w(y,T)$	Total implied variance $\sigma_{\text{imp}}^2 T$
$\sigma_{\text{loc}}(t,y)$	Dupire local volatility
$L(t,y)$	Leverage function
v_t; $(v_0, \theta, \kappa, \xi, \rho)$	Stochastic variance and its parameters
η	Mixing fraction (share of the vol-of-vol kept)
N, M, G, h	Particles, time steps, regression grid points, kernel bandwidth

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